

Corrected Pseudo-Label Self-Training for Monocular Pallet Pose Estimation under an Explicit Real-Supervision Budget

Technical manuscript for author review

Abstract—Synthetic training reduces the need for task-specific real annotations, but self-training can reinforce inaccurate target-domain landmarks. We examine whether correcting pseudo-label coordinates improves monocular pallet pose estimation when the student training conditions are otherwise matched. A frozen PoseFix-derived teacher, adapted using nine real images and 38 manually verified corners with synthetic replay, corrects a synthetic-pallet-pretrained model’s predictions. Raw and corrected students share 217 target images, supervision support, initialization, augmentation settings, synthetic replay, and 320 optimizer updates. On 66 directly verified visible development points, correction increases 10-pixel keypoint accuracy from 44 to 50 out of 66. On a repeatedly used, recording-disjoint development panel of 128 ordinary-plastic images, corrected versus raw student accuracy improves from 47.51% to 51.47%, while fixed-pipeline geometry-reference pose AUC improves from 0.33472 to 0.35902. The synthetic-only baseline obtains 49.14% and 0.33796. Both previously tested learning-rate settings and the available order repeats are retained. Tail error does not uniformly improve, and the pose reference is not independently measured physical ground truth. The evidence supports a scoped development result for corrected-target self-training, not zero-real-label learning, an unseen-session guarantee, or universal large-error recovery.

Index Terms—Pallet perception, monocular pose estimation, pseudo-labels, self-training, synthetic-to-real adaptation, evaluation provenance.

I. Introduction

An RGB pallet-pose estimator must localize geometric landmarks before recovering object position and orientation. Synthetic images can provide pallet-specific supervision, but real images introduce lighting, occlusion, and viewpoint shifts. A confident predicted box need not imply accurate corners. Consequently, treating every confident prediction as a training target can preserve or amplify localization errors.

The question in this work is deliberately narrower than developing an additional pose architecture: under a disclosed real-supervision budget, are corrected pseudo-label coordinates more useful than the original coordinates for student adaptation? A gain over the original synthetic-only model is insufficient to answer this question. It could instead result from additional optimizer updates, synthetic replay, or a different subset of target images. A useful comparison must hold those factors fixed and compare raw-target and corrected-target students directly.

We close this comparison using an existing matched experiment rather than introduce further modules after

inspecting development results. A frozen teacher supplies corrected coordinates, while the student is used without that teacher at inference. We distinguish four questions: label quality on the same verified points; corrected versus raw student performance; practical value over the synthetic-only baseline; and the population and metric scope of the evidence. The contributions are (i) an auditable matched-coordinate self-training comparison with explicit manual supervision, (ii) a joint 2D and fixed-pipeline 6D evaluation with complete denominators, and (iii) an evidence-scope analysis retaining order sensitivity, error tails, and failed examples. Self-training, pose refinement, and PnP are not claimed as new algorithms.

II. Related Work and Scope

STAC uses confident teacher detections as pseudo-labels and trains a student with augmentation-based consistency[1]. This motivates separating target quality from student training. Our experiment concerns pallet corner coordinates and matched support, and does not reproduce STAC’s object-detection benchmarks or claim its annotation-efficiency results.

PoseFix refines an input human pose from the image and synthesizes training errors using human-pose error statistics[2]. Our teacher is a pallet-specific adaptation of this family, not the original human benchmark. It uses pallet landmarks, a synthetic-trained prior, limited real manual coordinates, and synthetic replay. At self-training time the teacher is frozen; there is no human correction of each pseudo-label.

Self6D demonstrates synthetic-to-real pose adaptation with differentiable rendering and unannotated real RGB-D data during adaptation[3]. Our tested correction path instead consumes RGB, a predicted box, and predicted landmarks; it does not use a depth sensor or a rendering-based adaptation objective. Registered dimensions and camera intrinsics are used only by downstream geometry and the existing consistency filter.

BOP established evaluation that explicitly addresses object symmetries and pose ambiguity[4]; subsequent challenge reports distinguish datasets, tasks, and seen versus unseen object conditions[5]. Those distinctions motivate careful reporting here. Our normalized, symmetry-aware cuboid AUC is a repository-specific geometry-reference measurement, not a claim of BOP benchmark equivalence or state-of-the-art performance. None of these prior

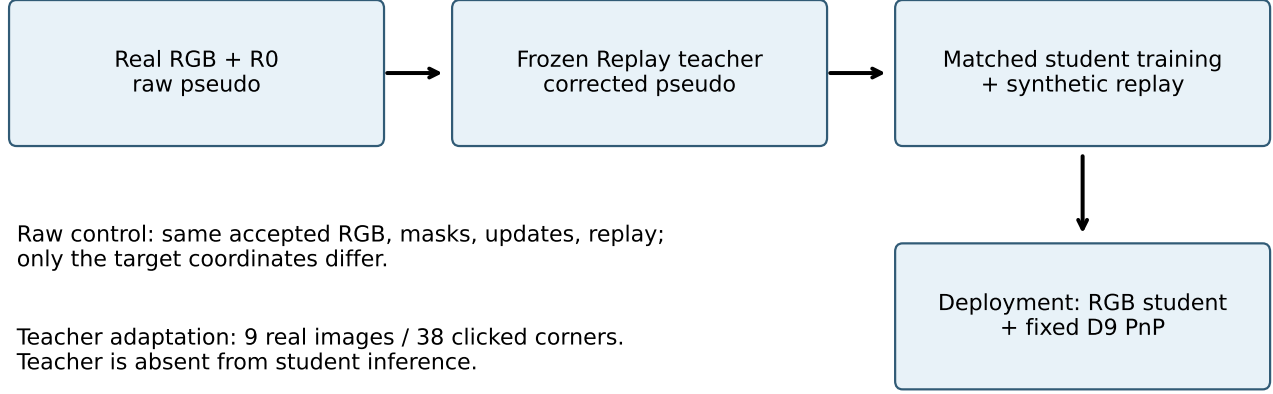


Fig. 1. The tested information paths. The teacher changes training targets, not deployed student architecture. Both students use the same accepted real images and target-support masks. The final geometry stage is fixed across arms.

papers’ numerical results is directly compared across incompatible datasets.

III. Method and Matched Intervention

A. Baseline and point contract

The baseline R0 is the existing YOLO26n pallet-pose checkpoint. “Synthetic-only” describes its pallet-specific fitting, not the provenance of all pretraining: upstream COCO-pose pretraining is documented. For each image, the fixed inference wrapper returns candidate boxes, scores, and nine landmarks. Corners 0–7 follow the registered cuboid convention; landmark 8 is the existing center. Instance selection and original-image coordinate restoration are unchanged across the comparison. Ground-truth boxes or poses do not choose an inference candidate.

B. Frozen correction teacher

The teacher is the existing RGB ResNet152 PoseFix-derived pallet network. It was initialized from the synthetic-trained PRIOR1 checkpoint and adapted for 300 updates using eight real and eight synthetic examples per update. The real term supervises only 38 verified manual corners in nine images: three ordinary-plastic night images and six wood-day images. Unknown and PnP-completed points and the center do not supply manual teacher supervision. Frozen batch-normalization statistics and trainable affine parameters follow the original protocol.

Teacher adaptation retains the existing heatmap cross-entropy plus coordinate loss and source replay objective. Real inputs alternate between R0 predictions and manually supported coordinates perturbed with predicted-box-scaled noise. We do not claim that synthetic and real errors have identical distributions. The original fixed final teacher checkpoint is reused without a new fit or development-selected displacement cap. At pseudo-label generation, it consumes only RGB, R0 points, and the predicted crop. It preserves the detection box, score, keypoint confidence, and center, and modifies corner coordinates.

C. Shared image selection and support

The ordinary-plastic candidate pool contains 1,000 real images outside the evaluation recordings. The existing raw confidence, horizontal-flip agreement, and leave-one-out (LOO) consistency checks retain 259 candidates; applying the frozen teacher and the existing post-correction LOO check retains 249. This pipeline does not imply that an accepted prediction is correct. No new geometric-shape, X-intersection, stability, or threshold sweep is introduced.

Both training arms use this same accepted set. A fixed replacement draw produces 512 real slots containing 217 unique images per epoch. For every raw/corrected pair, the box and trusted-keypoint mask are identical. The latter is the intersection of the original target-export validity masks; points outside shared support are ignored in both arms. Thus the controlled contrast concerns corrected versus raw coordinates conditional on a shared teacher-filtered selection policy. It does not estimate the effect of teacher-based selection itself, nor does the raw arm eliminate every influence of the manually adapted teacher.

Let $\tilde{\mathbf{y}}_i^r$ and $\tilde{\mathbf{y}}_i^c$ denote the raw and corrected targets and m_{ij} their common support. With the same augmentation A , real batch B_t , and synthetic replay batch S_t , the intervention is

$$\mathcal{L}_t^a = \mathcal{L}_{\text{pose}}(f_\theta(Ax_i), A\tilde{\mathbf{y}}_i^a; m_i) + \mathcal{L}_{\text{source}}, \quad a \in \{r, c\}. \quad (1)$$

This expression describes the existing pose objective and shared support, not a newly designed loss. Identical frozen detector branches retain the same ancillary detection terms. The only altered supervised targets are the real coordinate values.

D. Student training and deployment

Each student starts independently from the same R0 checkpoint. Only the pose branches and flow parameters are trainable; backbone, neck, box/class branches, and all running buffers remain exactly fixed. Each epoch

TABLE I

Matched coordinate intervention. Shared image membership is conditioned on the same raw-plus-refined filter acceptance, so this is not a comparison of different selection policies.

Contract	RAW and corrected
Initialization	Same R0 checkpoint
Trainable state	Pose branches and flow only
Real RGB	217 identical unique images
Real/synthetic exposure	2560 / 2560 each
Synthetic pool	512 identical images; no negatives
Supervised support	Identical raw/ref confidence intersection
Optimizer / updates	AdamW / 320
Learning rate	1e-5 main; 1e-4 sensitivity
Augmentation / seed	Same settings / 42
Checkpoint choice	Fixed final last.pt
Teacher supervision	9 images / 38 manual corners
Only changed target	Supervised pseudo coordinates

uses 512 real and 512 synthetic slots; five epochs with batch 16 give 320 updates and 2,560 exposures of each domain. The synthetic pool has 512 unique images and no negative images. Training uses AdamW, cosine decay, and the existing true-ignore keypoint implementation. The checkpoint is the fixed final checkpoint, never an epoch chosen by evaluation performance.

The historical experiment tested learning rates 10^{-4} and 10^{-5} with matched source-only, raw-target, and corrected-target arms. The latter rate was selected in the earlier reused development study. We disclose that selection, report both settings, and retain both available order repeats at the selected rate. Orders 43 and 44 change sample order; they are not three independent initialization seeds. Corresponding first-batch image-tensor hashes match within each raw/corrected repeat. This is a first-batch check, not a claim that a complete per-step tensor trace was recorded. Table I lists the controlled factors.

At deployment, only the student processes RGB. The teacher is absent. The final pose stage uses the same original D9 width/depth hypothesis selector for every arm and solves from corners 0–7. Landmark 8 remains part of the original selector residual; that known design choice is held fixed rather than altered to favor an arm. No trained model-specific router, reference-dependent oracle, or additional inference-time correction enters the main comparison.

IV. Evaluation Protocol

A. Populations and reference provenance

The main panel comprises 128 ordinary-plastic images from seven recording groups, with 29 Clean, 21 Moderate-occlusion, and 78 Severe-occlusion images. Its recording groups do not overlap the pseudo-label training pool or the teacher’s manual-training sessions, including recorded aliases. Nevertheless, this is repeatedly inspected development data, not independent confirmation. The historical full 194-image plastic panel included three teacher-training images; that aggregate is not substituted for the present comparison. The 300-image mixed-material panel is not pooled with these denominators.

Label-quality evaluation uses 66 directly verified visible corners in 16 images within the 128-image panel. The frozen FINAL_V2 reference retains only manually clicked, directly visible points after human metadata QA. Scoring uses fixed native point identity, without choosing a symmetry permutation. Missing points receive the native image diagonal as error. This is a selected, labeled development proxy for pseudo-label quality; it cannot establish the unknown accuracy of every accepted unlabeled adaptation image or the accuracy of hidden corners.

The full-panel legacy labels contain mixed point provenance. Their geometry-resolved poses combine image annotations, intrinsics, and registered dimensions and are not external physical 6D measurements. Because image references and pose references are related, agreement on both does not constitute two independent forms of metrological validation.

B. Metrics and fixed denominators

The main localization metric is PCK10, the fraction of all supported corners within 10 original-image pixels. PCK5 and PCK20, median error, P90, and errors above 20 pixels are also retained. These pixel thresholds are transparent operating tolerances for this dataset, not asserted universal pallet or robotics standards. Full-panel scoring uses one permitted whole-object symmetry permutation, not independent point-wise nearest matches. Fixed-identity results are retained separately. All 128 images contain detections, but only 120 meet the unchanged box-IoU matching gate of 0.5. Full-denominator PCK includes penalties for the remaining images over 985 supported corners; matched median/P90 uses 931 observed corners. Neither difficult frames nor failures are removed from PCK.

Final pose reporting includes coverage, width/depth-axis correctness, rotation and yaw error, translation error in centimeters, cuboid IoU3D, and normalized ADDsym AUC. The latter integrates the unchanged symmetry-aware error-threshold accuracy curve up to 0.1 object diameter. Failed poses count against AUC; every arm yields a pose for all 128 images here. An axis-correct hypothesis need not have accurate full rotation, and favorable AUC need not improve the median of every pose statistic.

The primary 2D contrast isolates coordinate learning; the common final pose pipeline tests its downstream value. Model-conditioned selectors and nondeployable reference oracles remain separate historical extensions. No new latency or operational-safety claim is inferred from these accuracy measurements.

C. Uncertainty and retrospective disclosure

We report paired per-frame changes, correct/incorrect corner transitions, every recording group, both learning-rate settings, and all order repeats. A 10,000-draw recording-cluster bootstrap describes PCK10 differences.

Corners are not treated as independent experimental replicates. With only seven reused recording groups and historical configuration search, these intervals are exploratory and not search-adjusted. The present closure fixes the scope before new re-evaluation, but is not a prospective preregistration of the original experiments.

V. Results

A. Q1: Does the teacher improve target quality?

On the same verified points, PCK10 increases from 44/66 to 50/66, and median error decreases from 7.144 to 5.489 pixels (Table II). The P90 and count of errors above 20 pixels also decrease. This supports improved visible-point quality on this proxy, not perfect correction or unmeasured accuracy on all training targets. Figure 2 includes both an improvement and a deterioration selected from the actual paired errors.

B. Q2: Does corrected-target training improve the student?

At the previously selected learning rate, corrected-target training raises PCK10 from 47.51% to 51.47% (Table III), a 3.96 percentage-point difference. The descriptive recording-bootstrap interval is [1.83, 6.46] percentage points. The paired mean scored frame error improves on 92 images, worsens on 28, and is unchanged on eight. At the 10-pixel boundary, 51 corners become correct and 12 become incorrect. The eight unchanged frame scores include detection-gate penalty ties; they are not necessarily identical predicted coordinates. No raw corner above 20 pixels becomes correct within 10 pixels in this main student contrast. The gain should therefore not be described as successful general large-error recovery.

The pose stage also improves under identical selection and solver code: ADDsym AUC changes from 0.33472 to 0.35902 (Table IV). This is a final-pipeline result, not an oracle choice among poses. The other historical learning rate and both order repeats retain a favorable corrected-versus-raw direction for pooled PCK10 and AUC (Table VI). They support consistency within the explored settings, not independent dataset replication.

Importantly, on the smaller manually reverified subset the two students tie at 43/66 PCK10, compared with R0's 44/66 (Table VIII). Corrected-student PCK20 and P90 improve there, but its median is slightly worse than the raw student's. Thus the full-panel PCK10 gain is not reproduced at that threshold on this visible-only reference. Teacher quality and student improvement are distinct measurements; the favorable teacher result cannot replace this student check.

C. Q3: What is added beyond synthetic-only R0?

R0 obtains 49.14% PCK10 and 0.33796 pose AUC, compared with 51.47% and 0.35902 after corrected-target self-training. The raw-target student is worse than R0 on those pooled metrics, so the result is not simply a benefit of any

self-training. The source-only update controls are included to expose effects of additional source optimization. No detector improvement is claimed: detector boxes, scores, and candidate order remain exactly unchanged.

There is no all-metric dominance. Matched-corner P90 is 40.902 pixels for R0, 41.958 for the raw student, and 42.085 for the corrected student. The corrected model's median rotation error and axis count do not surpass R0. AUC improves despite a worse median normalized ADDsym error, reflecting different regions of the error distribution. These are observable trade-offs, not omitted measurements.

D. Q4: Where does the result hold?

Table V retains all difficulty levels. Corrected versus raw PCK10 and AUC improve in all three strata at LR5, but corrected versus R0 Moderate PCK10 is lower. Severe pose AUC changes little relative to R0, and its median translation error remains worse. Recording-wise results are retained in Table VII; small groups must not be interpreted as precise new-domain estimates. The strongest claim is pooled and scoped to these ordinary-plastic development recordings.

VI. Limitations and Evidence Closure

First, the teacher uses real manual supervision. Our intervention is conditional on nine adaptation images and 38 corners, as well as a shared teacher-filtered training selection. We do not claim zero real labels, superiority to directly spending the same annotation budget on the student, or a measured reduction in human annotation time. The 66 verified evaluation points are additional human effort and never student-training targets. Generic pretraining and earlier development annotation/inspection also remain part of the full project history.

Second, the real adaptation data and evaluation recordings are disjoint, but the evaluation panel has been repeatedly used for method development. The LR5 setting was chosen in that history. Session-disjoint fitting does not undo that repeated inspection. The reserve provenance audit found no currently certified untouched, labeled confirmation panel; filename labels such as FINAL are not proof of independence. We therefore close a development-only manuscript without reporting an independent-confirmation number. A future untouched confirmation should be evaluated once with the locked pipeline, without using it to select the learning rate, filters, or checkpoint.

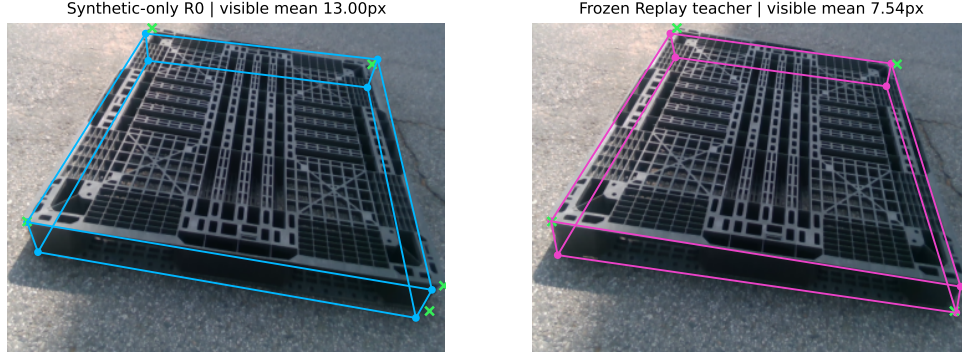
Third, the selected visible-point reference and mixed-provenance full-panel reference answer different questions. Neither proves hidden-corner accuracy or independent physical pose accuracy. The main result covers ordinary rectangular plastic, not wood or square green pallets. Related Clean19 occlusion-transfer, manually supervised hard-image, and model-specific selector studies involve different budgets or interventions and are treated only as extensions, not added to this causal contrast.

TABLE II

Pseudo-label quality on 66 verified visible points in 16 reused DEV images; fixed native identity. A labeled proxy, not accuracy measured on unlabeled adaptation images.

Output	Points	PCK5 %	PCK10 %	PCK20 %	Med px	P90 px	Above20
Synthetic-only R0	66	21/66 (31.82)	44/66 (66.67)	60/66 (90.91)	7.144	18.390	6
Frozen Replay teacher	66	30/66 (45.45)	50/66 (75.76)	63/66 (95.45)	5.489	13.759	3

eval_cad:1778653033056483584 | verified green crosses | corrected - raw -5.46px



eval_cad:1778653018878592768 | verified green crosses | corrected - raw +6.28px

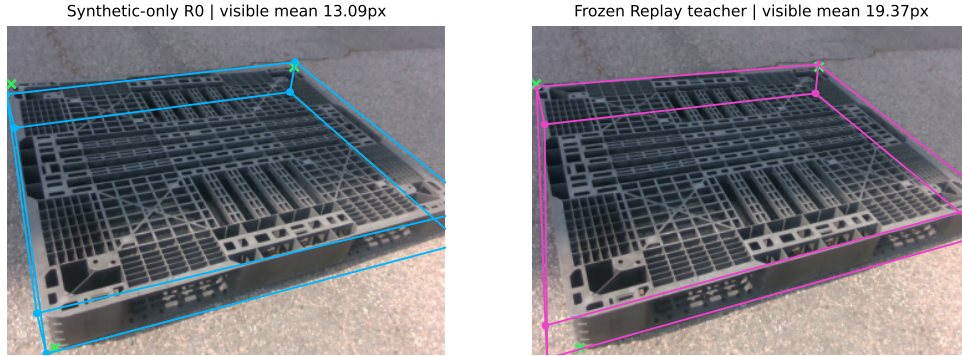


Fig. 2. Frozen raw versus corrected pseudo-label examples: one favorable and one unfavorable visible-point case. Green crosses are directly verified visible references. The actual images and predictions are reused, not synthesized; examples are selected post hoc for explanation and do not determine inference or aggregate membership.

TABLE III

Main 2D comparison on 128 images and 985 supported corners. PCK includes detection failures; medians and P90 are conditional on matched detection. LR5 was historically DEV-selected.

Arm	PCK5 %	PCK10 %	PCK20 %	Med px	P90 px
R0	20.81	49.14	72.69	9.565	40.902
RAW_LR5	20.30	47.51	72.59	9.961	41.958
REF_LR5	24.67	51.47	73.91	8.954	42.085

Finally, the pose-head-only matched experiment does not show that arbitrary whole-model fine-tuning or other teacher architectures will improve. P90 and severe errors remain unresolved. Figure 4 explicitly includes a worsened case and a low-change case. No additional architecture, loss, selector sweep, or annotation round is triggered to remove these limitations. The appropriate next action is manuscript review and, if required, independent confirma-

tion, rather than open-ended method search.

VII. Conclusion

With identical target images, supervision masks, initialization, augmentation settings, synthetic replay, and update budget, corrected pseudo-label coordinates improve the observed ordinary-plastic student’s pooled keypoint accuracy and fixed-pipeline pose AUC relative to raw pseudo-labels. The same metrics also improve relative to the synthetic-pallet-only baseline, under an explicit nine-image, 38-corner teacher-adaptation budget. The result is supported within reused development data and does not establish uniform tail recovery or independent physical-pose generalization. This scoped conclusion closes the central experiment without requiring every auxiliary metric to improve.

TABLE IV

Final 6D under the SAME frozen D9 selector and corner-only solver. Reference is geometry-derived, not independently measured physical pose.

Arm	Pose	Axis	R deg	Yaw deg	t cm	IoU3D	ADDsym AUC
R0	128/128	83/128	3.530	2.756	9.353	0.5865	0.33796
RAW_LR5	128/128	81/128	3.942	2.626	9.226	0.5808	0.33472
REF_LR5	128/128	82/128	3.766	2.550	9.186	0.5928	0.35902

TABLE V

All severity strata retained. Improved pooled accuracy does not imply improved tails or all pose statistics.

Group	N	Arm	PCK10 %	P90 px	ADDsym AUC	t cm
CLEAN	29	R0	65.94	17.485	0.71474	2.930
CLEAN	29	RAW_LR5	65.50	16.772	0.73053	2.818
CLEAN	29	REF_LR5	69.00	17.150	0.78126	2.218
MODERATE	21	R0	58.44	24.119	0.47955	5.907
MODERATE	21	RAW_LR5	55.19	24.087	0.47131	6.376
MODERATE	21	REF_LR5	56.49	23.008	0.51188	6.242
SEVERE	78	R0	40.37	53.725	0.15976	13.677
SEVERE	78	RAW_LR5	38.70	55.113	0.15079	14.692
SEVERE	78	REF_LR5	43.52	55.959	0.16087	14.431

Fixed128 reused DEV — no population or selector changes

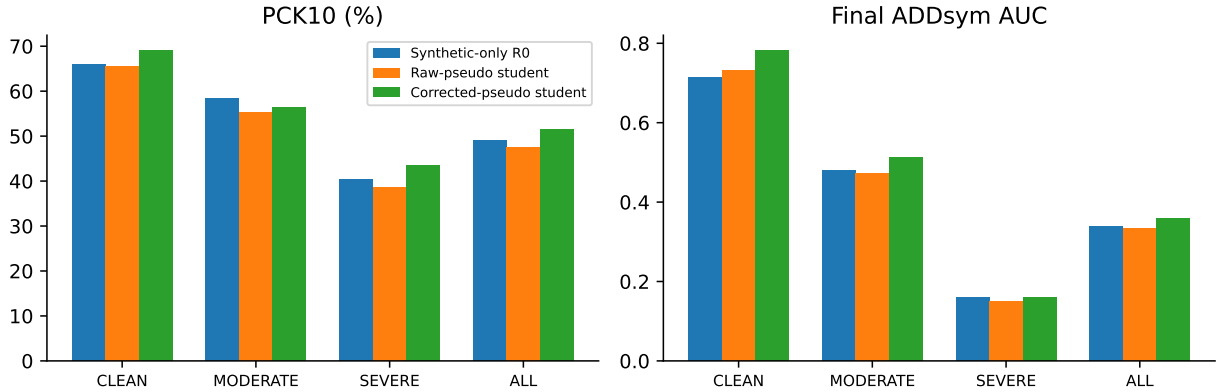


Fig. 3. All pre-existing severity strata, under the same evaluation contract. Pooled gains do not establish uniform tail-error recovery or independent generalization.

Appendix A

Sensitivity, Failure Cases, and Reproduction

Table VI contains all twelve existing student fits; no new fitting was performed in the original evidence-closure pass. A subsequently bounded, training-motivated budget experiment is reported separately below and does not replace that main comparison. Table VII retains every evaluation recording. The repository package provides checkpoint and source hashes, full optimizer and augmentation arguments, split and target-support contracts, common-selector predictions, numerical provenance for every table, and the

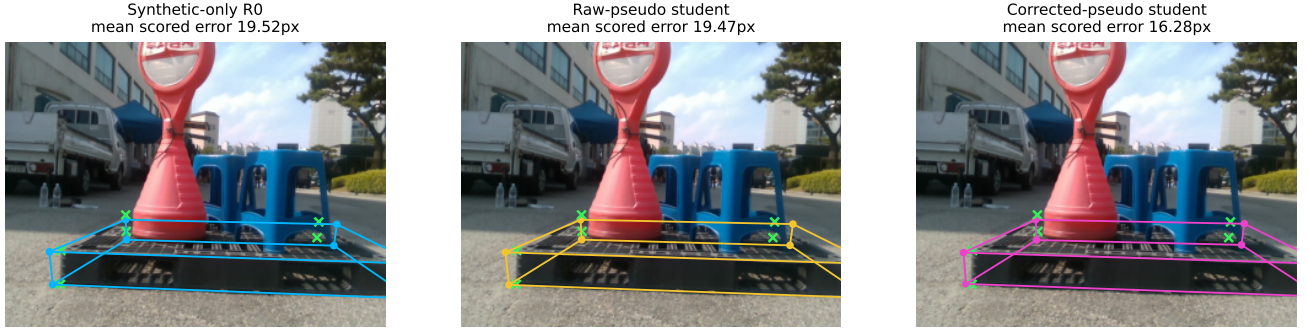
paired audit. Exact private RGB/annotation access is needed for full regeneration; hash manifests alone are not a public dataset release. Code and report location: <https://github.com/Canele452/pallet-6d-pose>, namespace pallet_selftraining_paper_closure_v1.

Appendix B

Post-Hoc Visible-Transfer Diagnosis and Bounded Budget Test

This analysis retains the original R0, raw LR5, corrected LR5, and nine-image/38-corner Replay teacher. No new

improved: eval_pallet07:1778652146612550912 | SEVERE_OCCLUSION | corrected - raw -3.19px
 Green: legacy reference (mixed provenance); lines: native 2D predictions, NOT PnP projection



worsened: eval_night09:1779449580573721600 | SEVERE_OCCLUSION | corrected - raw +1.27px
 Green: legacy reference (mixed provenance); lines: native 2D predictions, NOT PnP projection

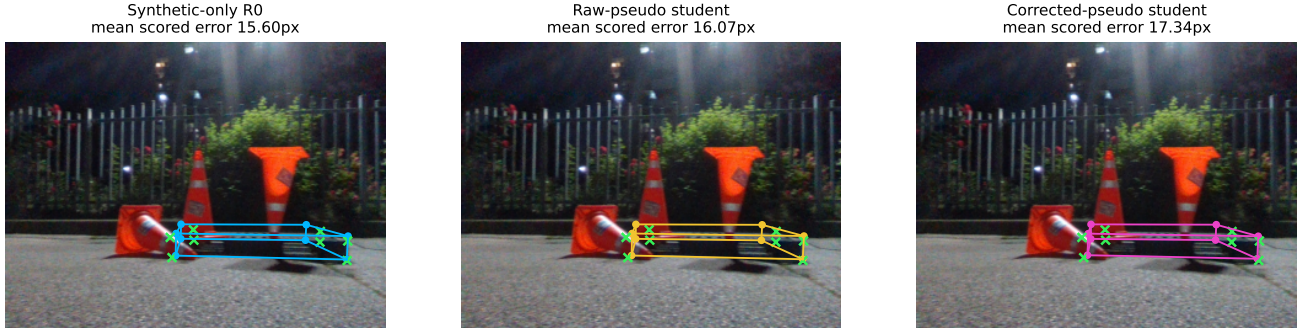


Fig. 4. Actual student examples with strong improvement (top) and deterioration (bottom) in mean scored corner error. The cyan/yellow/magenta native prediction overlays are R0/raw/corrected students, respectively; green crosses are legacy references of mixed provenance. These are raw 2D predictions, not PnP reprojections. Full-panel error scoring uses the frozen whole-object symmetry contract; image lines retain native identity.

TABLE VI

Complete historical grid and order sensitivity. SYN is additional source-only training. ORDER43/44 change data order, not initialization seed. No best-repeat selection.

Arm	PCK10 %	P90 px	ADDsym AUC
R0	49.14	40.902	0.33796
SYN_LR4	47.41	42.356	0.33443
RAW_LR4	47.31	42.343	0.34073
REF_LR4	50.76	42.569	0.35311
SYN_LR5	48.63	41.776	0.33895
RAW_LR5	47.51	41.958	0.33472
REF_LR5	51.47	42.085	0.35902
SYN_ORDER43	48.93	41.762	0.33945
RAW_ORDER43	48.32	41.808	0.33799
REF_ORDER43	51.78	42.653	0.36100
SYN_ORDER44	48.83	41.760	0.33994
RAW_ORDER44	47.72	41.951	0.33684
REF_ORDER44	51.27	42.294	0.35880

TABLE VII

Every reused recording group, including small or adverse groups. Percentages are descriptive, not independent frame trials.

Recording	N	Raw PCK10	Corr PCK10	Raw AUC	Corr AUC
REC_007	33	47.71	49.24	0.08383	0.09958
REC_021	18	39.71	45.59	0.80000	0.81567
REC_022	16	32.79	36.89	0.22866	0.24034
REC_025	27	38.07	44.16	0.36820	0.38743
REC_027	12	26.09	36.96	0.14642	0.12592
REC_041	10	71.25	71.25	0.23680	0.31110
REC_044	12	96.88	96.88	0.66275	0.75483

TABLE VIII

Verified-visible student sensitivity: both students tie at PCK10; the legacy-panel gain is not reproduced at this threshold on the small manually reverifed subset. Same66 points, fixed identity.

Arm	PCK10	PCK20	Median px	P90 px
R0	44/66	60/66	7.144	18.390
RAW_LR5	43/66	60/66	7.046	19.574
REF_LR5	43/66	63/66	7.097	17.343

RGB, manual coordinates, teacher fit, selector, reliability filter, or refiner architecture was introduced. The 66 points were already inspected development data; neither freezing predictions again nor their absence from student training makes them an independent test.

A. Why the verified-point result tied

Across all 66 fixed-identity points, the original raw-to-corrected student contrast has three entries into and three exits from the 10-pixel correct set. The net changes

least_changed: eval_night08:1779449483432542720 | SEVERE OCCLUSION | corrected - raw +0.00px
 Green: legacy reference (mixed provenance); lines: native 2D predictions, NOT PnP projection

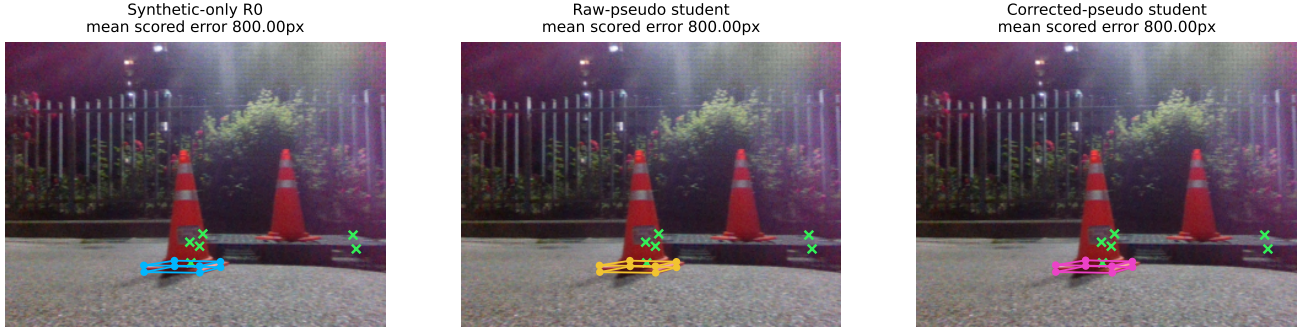


Fig. 5. A low-change scored case retained alongside favorable and unfavorable examples. Identical detection penalties can yield an unchanged score even if point coordinates move; no inference correction was chosen using these references. The Markdown report contains all 13 paired example images.

are +2 Clean, -2 Moderate, and 0 Severe. Continuous error improves at 34 points and worsens at 32, with a mean difference of -0.461 pixels. Equal PCK10 therefore does not establish equivalent predictions. Each student has ten points where only the teacher is correct and three where only the student is correct; teacher correctness at an evaluation point is not evidence that the point was a training target.

Using identical predictions, point IDs, native coordinates, and missing-value handling, the same 66 points score 45/66 raw versus 44/66 corrected under the legacy reference, and 43/66 versus 43/66 after verification. The median legacy-verified reference displacement is 2.828 pixels (P90 5.374). Thus the corrected student’s full-panel advantage was already absent on this selected subset before reference replacement. The full-panel/subset difference cannot be attributed entirely to annotation error. No reference was automatically repaired, and PCK10 remains the principal visible-point threshold.

B. Training-target following and locked intervention

The actual real-image occurrence list contains 217 unique images and 512 slots per epoch. Among 1,627 commonly supported corners, mean residual to the corrected target is 4.078 pixels for the raw student and 3.070 for the corrected student. For the 1,492 corners with target displacement at least one pixel, 82.64% of student differences have positive projection onto the target-correction direction; the median scalar projection fraction is 0.319. This describes partial target following, not physical accuracy: the unlabeled training targets are not ground truth. Unique-image-uniform and occurrence-weighted summaries, center-only results, and recording/corner/support strata are retained separately in the repository.

The diagnostic uses unaugmented native images and highest-confidence instances, never target-nearest candidate replacement. Saved padded training RGB is bit-exact with the original single 100-pixel reflection padding, and

TABLE IX

Bounded post-hoc budget test, not a replacement for the 320-update main experiment. TRAIN residual is to the corrected pseudo target, not to real ground truth. Full-panel PCK10 retains all 985 supported corners.

Arm / updates	PCK10 /66	Full (%)	AUC	Fit px
Raw / 320	43/66	47.513	0.33472	4.078
Corrected / 320	43/66	51.472	0.35902	3.070
Raw / 640	44/66	47.614	0.33681	4.032
Corrected / 640	44/66	51.168	0.36189	2.994

the target inverse transform agrees within 10^{-5} pixels. Recomputed R0 outputs agree with raw targets to below 10^{-6} pixels on average. There is no exhaustive augmented-tensor cache. Training losses mix source and real examples, use different area/support reductions, and are not native pixel residuals. The original coordinate loss decreases overall and in its final interval but is not monotonic. Remaining target residual and late loss decrease motivated one budget-only hypothesis test; contradictory pseudo supervision, augmentation effects, replay, and frozen representation remain competing explanations.

Both new students start from the identical R0 and run 640 optimizer updates, preserving the original first 320-update learning-rate and end-to-end loss-weight schedules. The following 320 updates hold the actual final-prefix learning rate, $1.859423525 \times 10^{-6}$. Simply changing the cosine horizon to ten epochs would also change the prefix and was not used. For each arm, first-prefix training-loss/LR CSV values and saved EMA weights reproduce the original run exactly; all 747 protected tensors remain exact. This is not a claim of an exhaustive historical training-tensor audit. All images, boxes, common support, order, augmentation settings, and optimizer parameters remain matched, with only raw/corrected target coordinates differing. Both fixed final checkpoints are used. Their predictions and unchanged D9 poses were hash-locked before reference scoring.

C. Mixed result and stopping decision

Both students gain one verified correct point, so the corrected-versus-raw tie remains (Table IX). Corrected 640 versus corrected 320 has one entry and no exit at PCK10, while corrected versus raw at 640 has two entries and two exits. Corrected PCK5/PCK20 remains 26/66 and 63/66. Its mean verified error decreases from 10.082 to 10.040 pixels and P90 from 17.343 to 17.005, but median error worsens from 7.097 to 7.120. The corrected training-target residual falls by only 0.076 pixels. These small changes do not establish that additional optimization resolved visible transfer.

On the unchanged full panel, corrected 640 versus corrected 320 loses three net correct corners (507/985 to 504/985; two entries, five exits), while pose AUC rises from 0.35902 to 0.36189. Matched P90 worsens from 42.085 to 42.160 pixels. Under the same 640-update budget, corrected still exceeds raw by 3.553 PCK10 percentage points and 0.02509 AUC, compared with 3.959 points and 0.02429 at 320 updates. Thus the intervention’s effect and the corrected-target added value are different contrasts: the former is mixed, and the latter remains favorable on this reused legacy panel but not superior at verified PCK10. Detection remains 128/128, matched detection 120/128, and pose coverage 128/128 for all arms. The original 32-image synthetic validation role and all severity/recording summaries are retained without calibration or checkpoint selection.

This single paired experiment does not uniquely identify the remaining cause or rule out every possible training budget. It provides no independent-session, hidden-point, or physical-6D validation. We preserve the original main results and close method development without a second intervention, additional labels, or a new uncertainty module. Full contrasts, damaging and low-change examples, hashes, and reproduction instructions are provided in the repository namespace `pal-let_visible_transfer_closure_v1`.

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